Software Test Design (STD)

Emily Myaskovski | Ido Katz

1. Introduction

This Software Test Design (STD) document provides detailed test cases for verifying the functionality of the AI-Based Real-Time Defect Detection in 3D Printing project. Based on the approved Software Test Plan, this document outlines specific preconditions, sequential test steps, and expected outcomes to validate the system's edge computing operations, computer vision inferences, user interface dashboard, alerting engine, and system resilience.

2. Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
TC-1.1	Click "Start Stream"	Camera is connected via USB.	1. Connect the camera to the Raspberry Pi. 2. Navigate to the Live Monitoring Dashboard. 3. Click the "Start Stream" button.	The system detects the camera, streams frames to the canvas.	The system detects the camera, streams frames to the canvas.
TC-1.2	UI Dashboard: Stop Capture	Active video stream.	1. Click "Stop Capture" button.	Active video capture session stops, and monitoring stream terminates safely.	Active video capture session stops, and monitoring stream terminates safely.
TC-1.3	Handle sudden camera disconnection	System is actively streaming.	1. Ensure active video feed is running. 2. Physically disconnect the USB camera.	The system does not crash. A managed error is printed in the logs: "Camera disconnected".	The system does not crash. A managed error is printed in the logs: "Camera disconnected".

TC-1.4	Multiple clicks on "Start Stream"	None.	1. Click the "Start Stream" button multiple times rapidly.	Additional clicks are ignored. Duplicate camera threads are not spawned.	Additional clicks are ignored. Duplicate camera threads are not spawned.
TC-2.1	Detect Spaghetti Defect	Present a tangled plastic object to the camera.	1. Place a spaghetti-like plastic tangle within camera view. 2. Observe the inference bounding boxes and labels.	The model draws a bounding box around the anomaly, classifying it as 'Spaghetti' with >85% confidence.	The model draws a bounding box around the anomaly, classifying it as 'Spaghetti' with >85% confidence.
TC-2.2	Detect Stringing (Resolution challenge)	Present a print with fine, semi-transparent strings.	1. Place a 3D print exhibiting stringing artifacts in view. 2. Observe the detection overlay.	The model successfully isolates the strings as 'Stringing' without confusing them with the printer's background.	The model successfully isolates the strings as 'Stringing' without confusing them with the printer's background.
TC-2.3	Multiple Defects	An image containing both spaghetti and stringing in different areas.	1. Input the multi-defect image into the system. 2. Verify bounding box rendering.	The algorithm identifies both entities independently, drawing two separate bounding boxes with their respective labels.	The algorithm identifies both entities independently, drawing two separate bounding boxes with their respective labels.

TC-3.1	Real-time Confidence Threshold adjustment	Stream a video containing a borderline defect (e.g., 60% confidence).	1. Adjust the threshold slider to 70%. 2. Adjust the threshold slider down to 50%.	At 70%, the alert is cleared. At 50%, the alert is triggered instantly.	At 70%, the alert is cleared. At 50%, the alert is triggered instantly.
TC-3.2	Visual Alert Trigger	A spaghetti defect is detected above the threshold.	1. Trigger a high-confidence defect detection event. 2. Observe the dashboard UI updates.	The UI displays a red warning banner, and the defect counter increments by 1.	The UI displays a red warning banner, and the defect counter increments by 1.
TC-3.3	"Clear Alerts" functionality	Active alerts and a non-zero counter are present on the dashboard.	1. Click the "Clear Alerts" button. 2. Verify the state of the UI panels.	Clicking the button clears the history, resets the counter to zero, and restores the UI to a clean state.	Clicking the button clears the history, resets the counter to zero, and restores the UI to a clean state.
TC-4.1	Sudden network disconnection during monitoring	The system is actively monitoring and streaming data to the UI.	1. Disconnect the client device from the network. 2. Observe the UI and backend logs.	The API on the Raspberry Pi does not crash. The UI displays a "Connection Lost" warning and attempts to reconnect.	The API on the Raspberry Pi does not crash. The UI displays a "Connection Lost" warning and attempts to reconnect.
TC-4.2	Recovery after network restoration	The system is in a "Connection Lost" state (post-test 4.1).	1. Restore network connectivity on the client device. 2. Observe dashboard behavior without refreshing.	The UI automatically reconnects to the API, the video stream resumes, and alert data synchronizes without requiring a manual refresh.	The UI automatically reconnects to the API, the video stream resumes, and alert data

					synchronizes without requiring a manual refresh.
TC-4.3	Power loss and hard reboot	System is operating normally.	1. Disconnect power from the Raspberry Pi. 2. Reconnect power and wait for boot sequence. 3. Check the dashboard via browser.	The OS boots, background services (FastAPI and YOLO script) start automatically, and the system returns to an active monitoring state without user intervention.	The OS boots, background services (FastAPI and YOLO script) start automatically, and the system returns to an active monitoring state without user intervention.
TC-5.1	Low light to complete darkness	Camera is pointed at the printer.	1. Turn off room lighting completely. 2. Observe inference behavior under high digital noise.	The system does not crash. No false positive defects are generated due to digital camera noise.	The system does not crash. No false positive defects are generated due to digital camera noise.
TC-5.2	Camera lens obstruction	The system is running actively.	1. Place an opaque object directly covering the camera lens. 2. Observe the video feed and inference engine status.	The feed shows a black/obstructed screen. The AI algorithm does not crash and does not classify the obstruction as a "defect".	The feed shows a black/obstructed screen. The AI algorithm does not crash and does not classify the obstruction as a "defect".

TC-5.3	Focus loss / extreme blur	The system is running actively.	1. Move or adjust the camera so the printer is completely out of focus. 2. Observe alert generation.	No false positives are generated, and the system continues to function.	No false positives are generated, and the system continues to function.
TC-6.1	Auto-Purge 30-Day Retention Policy Execution	Database contains alert and detection records older than 30 days.	1. Start the FastAPI server to trigger the background purge task. 2. Inspect the SQLite database and storage directory.	All database records with a timestamp older than 30 days are automatically deleted. Newer records remain intact. The system continues monitoring without crashing.	All database records with a timestamp older than 30 days are automatically deleted. Newer records remain intact. The system continues monitoring without crashing.
TC-7.1	UI Dashboard: Feed Canvas & Overlays	Active video stream with defects.	1. Observe the real-time video stream.	Canvas displays AI-generated overlays: Bounding boxes, Confidence labels, and Detection status indicators.	Canvas displays AI-generated overlays: Bounding boxes, Confidence labels, and Detection status indicators.
TC-7.2	UI Dashboard: Feed Controls	Active video stream.	1. Click "Zoom In" button. 2. Click "Snapshot Capture" button. 3. Use Display/Overlay Adjustment.	Stream zooms in, snapshot is captured and saved, and overlays adjust accordingly.	Stream zooms in, snapshot is captured and saved, and overlays adjust accordingly.

TC-7.3	UI Dashboard: Status & Alerts	System monitoring is active.	1. Check Recent Alerts Panel. 2. Check Real-Time Status Indicator.	Panel displays live events/alerts with timestamps/confidence. Status shows Printing / Monitoring Active / Normal.	Panel displays live events/alerts with timestamps/confidence. Status shows Printing / Monitoring Active / Normal.
TC-8.1	Settings: Confidence Threshold Update	Navigate to System Settings.	1. Move the Confidence Alert Threshold slider to a new value (e.g., 75%). 2. Click 'Save Configuration'.	The percentage value updates on screen with matching color. A success notification appears, and the new threshold is applied to the live inference engine.	The percentage value updates on screen with matching color. A success notification appears, and the new threshold is applied to the live inference engine.
TC-8.2	Settings: Save & Reset Actions	Modified system settings.	1. Click "Save Configuration". 2. Click "Reset to Default".	Current parameters are saved. Reset restores all settings to default values.	Current parameters are saved. Reset restores all settings to default values.
TC-9.1	History: Export & Refresh	Alerts & History page, data exists.	1. Click "Export CSV". 2. Click "Refresh Data".	Alert history exports into CSV format. Alerts/monitoring data refreshes on screen.	Alert history exports into CSV format. Alerts/monitoring data refreshes on screen.
TC-9.2	History: Filters	Alerts & History page.	1. Apply Defect Type, Confidence, and Time Range filters.	Table filters alerts by category, confidence level, and selected time ranges.	Table filters alerts by category, confidence level, and selected time ranges.

TC-9.3	History: Event Log Table & Image View	Alerts & History page with at least one event.	1. Observe Detailed Event Log Table. 2. Click the "View Image" link/button.	Table displays Timestamp, snapshot image, Defect type, Confidence. Clicking “View Image” opens the full snapshot image in a new tab.	Table displays Timestamp, snapshot image, Defect type, Confidence. Clicking “View Image” opens the full snapshot image in a new tab.
TC-10.1	Preprocessing Module	Video frame inputted.	1. Run frame through preprocessing module.	Resolution scaling strictly converts frame to 640x640. Color channel maps BGR to RGB.	Resolution scaling strictly converts frame to 640x640. Color channel maps BGR to RGB.
TC-10.2	Inference Module (Core AI)	Preprocessed frame ready.	1. Load custom weights (best.onnx). 2. Execute prediction function.	Model classifies 3 target classes (Normal, Spaghetti, Stringing) and extracts bounding box coordinates for UI rendering.	Model classifies 3 target classes (Normal, Spaghetti, Stringing) and extracts bounding box coordinates for UI rendering.